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ORIGINAL ARTICLE

A novel augmentation strategy for credit scoring modeling

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Abstract
In last years, social lending platforms have been increasingly used as virtual environments where borrowers can directly interact with lenders without any intermediary. As a result, a reliable credit scoring strategy, i.e., assessing whether a client is able to fully repay a loan, became of utmost importance to reduce the risk of not repaying the lenders. In this context, machine learning tools are being increasingly adopted to design automatic credit scoring systems but the data imbalance problem still penalizes their predictive performance, i.e., the greatest majority of clients can afford the repayment and learning to classify "bad" borrowers depends on few instances where the loan was not paid back. In this paper, we target the data imbalance problem and propose a novel data augmentation strategy to improve the predictive performance of credit scoring models. The proposed methodology performs data augmentation by injecting synthetic instances in the dataset generated along the decision boundary of the decision model. We assessed the effectiveness of the proposed augmentation strategy on a million-scale dataset from Lending Club, the largest Social Lending platform, and found that it improves the performance of several classification models, also in comparison to other state-of-the-art approaches.

Keywords Credit score prediction · Supervised learning · Explainable artificial intelligence · Data augmentation

1 Introduction

Credit technology has undergone an unprecedented transformation with the advent of the financial technology (*Fintech*) revolution [9] and data science represents one of the most promising fields to understand the evolution of the financial market. Furthermore, the financial crisis in 2008 has determined an increased interest of researchers and practitioners in the stability of the financial system, which, in turn, has been significantly impacted by the action of the bank runs[1].

To support the continuous need to raise funds, Internet financing has emerged as an important alternative to banks where individuals or small and medium-sized enterprises (SMEs) borrow money using online trading platforms [47]. In particular, the rise of new services, named *Social Lending platforms* [1, 41], enables virtual interaction among borrowers and lenders without the intervention of any intermediary [18]. Concretely, these platforms provide individuals and SMEs with alternative credit options by offering better return rates to lenders and easy access to credit at affordable costs to borrowers who may have limited access to banks [19, 52].

However, the low access threshold [35], in combination with shorter loan periods [37], increases the risk of clients not repaying the loans they request. Consequently, credit scoring becomes an important financial tool for lenders to identify preventively "bad" borrowers who cannot fully repay their liabilities [57]. In addition, a lack of lenders' experience and missing or uncertain information about borrowers' credit history can increase risks in social lending platforms requiring accurate credit risk scoring. Hence, loan evaluation has become a critical tool for credit

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[1] *Bank runs* has been defined as "a shift in expectations, which could depend on almost anything, consistent with the apparently irrational observed behavior of people running on banks" [14].

risk assessment to reduce the default rate and increase lenders' returns [38]. In this context, Fintech innovations, powered by modern artificial intelligence tools, can provide semi-automated systems to price and underwrite loans [5, 42, 59] as well as to perform reliable and fair credit scoring.

Concretely, from a machine learning perspective, credit scoring is a classification problem where the model needs to decide whether a request for a loan should be accepted based on data related to the borrower (e.g., the employment status, the annual salary) and the characteristics of the loan (e.g., the interest rate, the time of repayment). Various machine learning techniques have been applied to the prediction of repaying a loan in the past few decades [13, 36, 57]. In particular, the availability of several customer data (i.e., credit history, demographic, and financial information) represented the foundation for designing machine learning models to evaluate the default risk associated with granting a loan to each borrower [7]. Different studies have been designed for investigating the main features offered by these platforms such as [16] and [39], that evaluate their relevance using logistic regression and predicting borrower status through random forest classifier, respectively. More recently, the focus has also shifted toward end-to-end classification pipelines built with modern deep sequential neural networks (e.g., bidirectional long short-term memory) that jointly learn a power vector representation [44] for time-series data and the respective credit score prediction function.

Nevertheless, the effectiveness of the above-mentioned approaches is threatened by the data imbalance problem that all credit scoring applications need to face, i.e., "bad" loans only take a small proportion of the whole dataset. In other words, the automatic learning process can rely on a few examples of non-repaid loans and can be confused by several examples of successfully repaid loans. While learning from an imbalanced dataset is a well-studied problem in several applications of machine learning [21, 25, 31, 51, 53], most of the approaches for credit risk scoring leverage classic sampling or ensemble-based strategies [22, 24, 40, 41].

In this paper, we target the data imbalance problem and propose a novel data augmentation strategy to improve the predictive performance of credit scoring models. Specifically, we leverage a novel eXplainable Artificial Intelligence (XAI), named *CASTLE* (Cluster-Aided Space Transformation Local Explanation) [29], which provides clustering-based rules to explain the model's knowledge and an explanation vector whose direction can be followed to support a single prediction of the model. In particular, the proposed augmentation strategy leverages this direction to generate synthetic instances which are then added to the training set.

Our experiments on a million-scale dataset from Lending Club, the largest Social Lending platform, prove the effectiveness of the proposed augmentation strategy in comparison to other state-of-the-art approaches. Overall, we found that our method improves the performance of several classification models and, from a broader perspective, we discovered that generating new instances along the decision boundary of the classification model always benefits the performance of the credit score model.

The remainder of this work is structured as follows: Sect. 2 provides a literature analysis for both credit scoring prediction and XAI tools in this domain while the fundamentals of *CASTLE* have been described in Sect. 3. Section 4 describes the proposed methodology integrating *CASTLE* for generating synthetic instances to improve the effectiveness in dealing with credit risk scoring. Finally, the evaluation made on a real-world Social Lending platform has been discussed in Sect. 5, whose results are discussed in Sect. 6. Finally, Sect. 7 summarizes the obtained results and suggests several areas for future work.

2 Related work

Identifying bad borrowers is a relevant task for different analyses (i.e., fraud detection [10, 17, 33]) and credit evaluation [30]. In particular, the evaluation of possible borrowers is the aim of the credit scoring task in order to reduce risks related to non-repayment of loan [11, 15].

Nevertheless, different challenges (i.e., lack of lender history or unbalanced dataset) are still faced in identifying bad customers, which can lead to huge financial losses [27].

2.1 Credit scoring approaches

The financial crisis that happened in 2008 has focused the attention of several researchers and practitioners on *credit scoring*, which has become a key point in credit risk management [3, 46]. In particular, [1] provided a literature review about Social lending from 2008 until 2020 underlining how it has become an emergent and fast-spreading phenomenon in the context of digital finance.

Pioneering approaches for automatic credit scoring [23, 55] mainly relied on generating specific rules for evaluating credit risk. While Hayashi [23] assessed credit risk by extracting rules through a hierarchical, recursive consideration of discrete variables prior to analysis of continuous data, Soui et al. [55] produced a set of rules through a credit risk evaluation model based on a search-based optimization problem. Nevertheless, the large amount of data available in this domain poses several challenges to the use of both techniques due to different

issues in generating rules, including computational complexity and semantic relevance.

Nevertheless, the credit risk scoring models are strongly affected by the lack of lenders' experience as well as the need to deal with missing or uncertain information about borrowers credit history [40].

Machine learning approaches In the last years, several studies have used artificial intelligence models [28, 50] for evaluating borrowers with the aim to deal with credit scoring, showing promising results. Zhou et al. [63] conducted a study about four different gradient boosting methods, whose parameters are optimized by the Bayesian approach, to model the credit risk of borrowers in P2P platforms. Li et al. [32] developed the eXtreme gradient boosting (XGB) method to deal with credit evaluation task, applying it to the personal loan scenario based on the open data set from the Lending Club Platform in USA. Another approach based on XGB has been designed by Liu et al. [34] to linearize and transform the original features into a high-dimensional sparse feature matrix, on which a graph-based neural network, named *forgeNet*, has been applied for predicting the credit risk. Kim et al. [26] aimed to improve the default prediction on social data through an ensemble semi-supervised learning method that combines label propagation and transductive support vector machine (TSVM), whose results have been merged on the basis of the Dempster–Shafer fusion method. A genetic algorithm based on a profit score optimization objective, which considers actual and potential returns, has been proposed by Ye et al. [60] for optimizing the combination of decision trees in random forest. Most recently, [44] has proposed a deep learning framework based on stacked bidirectional long short-term memory models (BiLSTMs) in order to improve predictive performance. Despite the increasing effort toward improved predictions, there are still challenges related to the highly unbalanced settings of the credit score problem, i.e., most of the times loans are paid back with success with just a small number of instances where the borrower cannot return money.

Ensemble learning Other approaches aim to deal with the credit scoring by making an ensemble of different classifiers (see [54, 61] for instance). The former is based on a voting-based outlier detection for supporting an ensemble model while gradient-boosting decision trees have been applied for credit risk assessment by combining based on a combination strategy called distance-to-model. In turn, a bagging supervised autoencoder classifier (BSAC) has been designed by Abdoli et al. [2] for jointly exploiting the superior power of supervised autoencoders and representation learning in classification and the Bagging mechanism to handle the irregularities in feature space. Runchi et al. [49] proposed the ensemble model based on logistic regression working in three phases: it, firstly, makes pre-processing operation on dataset which is, successively, sampled by using difference imbalance ratio on which sub-models are trained. Finally, each model is evaluated on the validation set for computing its weight in the ensemble for credit scoring.

Benchmark The pervasive use of machine learning in Fintech [9, 43] prompted researchers to apply them to credit scoring by designing several benchmark studies. In 2018, Namvar et al. [41] performed an empirical comparison of well-known machine learning-based classifiers on the basis of different sampling strategies. Other benchmarking studies have been proposed in [40, 58], where the former compared the effectiveness performance of 10 different classification models while the latter investigated the effectiveness of the most used machine learning-based classifiers, also evaluating the explainability of the three best models through different eXplainable artificial intelligence (XAI) tools.

eXplainable AI (XAI) Currently, in the context of credit scoring, several intepretability techniques have been applied to explain the predictions of the classification models [8, 12, 48, 56] as well as to assess the fairness of the final decisions [6].

On the contrary, we leverage XAI tools to improve the performance of the classification process with respect to the state-of-the-art approaches that rely on classical sampling strategies [40, 41]. Although they have shown their applicability in different domains [40, 62], they suffer from multiple limitations that may make results nongeneralizable [4]. Specifically, we propose a novel augmentation methodology integrating CASTLE [29] to generate synthetic instances to deal with class imbalance during the training process.

3 Theoretical background

The data augmentation technique proposed in this work relies on the eXplainable AI (XAI) framework CASTLE ([29]). Specifically, given our ML models, we will consider the set of misclassified samples (i.e. False positives and False negatives), and input them to CASTLE to obtain the corresponding *explanation vector*, i.e. a vector that tells us how the input sample should be perturbed so as to increase its probability of being correctly classified (*supporting direction*) or decrease it (*opposing direction*). In the following, we will provide a theoretical background on CASTLE and the computation of supporting/opposing directions.

3.1 Cluster-aided space transformation for local explanations (CASTLE)

CASTLE aims to provide a global system explanation by preserving the model-agnostic property of the local interpretable model-agnostic explanations (LIME) [45]. In particular, it provides both a global model knowledge and a local explanation by describing clusters as rules and using clusters' representative points (*pivots*) to make a space transformation.

CASTLE works in two main phases, described as follows:

- *Preparation phase*: first, CASTLE aims to find common behaviors and peculiarities among instances by applying the PureDBSCAN clustering algorithm able to find clusters with (1) *high purity*, i.e. instances pertaining to the same class should belong to the same cluster, (2) *high coverage*, i.e. the cluster covers as many instances as possible, and (3) *low overlap*, i.e. different clusters should not contain the same instances. At the end of the preparation phase, CASTLE chooses *pivots*, i.e. one representative point for each cluster.
- *Explanation phase*: in this phase, CASTLE performs a space transformation which maps the input instances into a k-dimensional space, where k is the number of pivots and each value indicates the similarity score between the input point and the corresponding pivot. Then, the same process of perturbation and transparent model fitting as LIME is followed so as to obtain a score for each pivot indicating its influence on the prediction of the model.

3.2 Supporting/opposing directions

The output of the transparent model fitted by CASTLE on transformed data can be described as the result of the combination of attractive and repulsive forces that influence the target instance's prediction. Specifically, each pivot p exerts a force on the target instance, with the direction and magnitude of this force determined by the sign and magnitude of the pivot's weight. A positive weight (i.e. one that positively contributes to the prediction) corresponds to an attractive force, while a negative weight corresponds to a repulsive force.

The contribution of each pivot to the prediction of the target instance is a function of both its weight and its proximity to the target instance. As the distance between the target instance and a pivot with a positive weight decreases, the likelihood of the predicted class increases and this likelihood increases even more if the weight of the pivot is large. By considering these forces as vectors and computing their sum, the resulting explanation vector is directed toward the region of the feature space that is most supportive of the prediction, and its magnitude is equal to the prediction itself. Additionally, the opposite direction of the explanation vector represents the area of feature space along which the likelihood of the target class decreases, providing insight into the opposing direction to the prediction.

4 Methodology

Figure 1 shows the overall flow of the methodology proposed for credit risk scoring, starting from data crawling from the largest social lending platform (*Lending club*[2]) until to the borrower estimation. In particular, our methodology relies on two main components: data processing operations (see Sect. 4.2) and machine learning-based classification (see Sect. 4.3), in which we integrate *CASTLE* XAI tool in the classification process for data augmentation.

4.1 Task definition

In the last decade, researchers have designed different approaches in order to deal with credit risk scoring, whose aim is to manage credit risk exposure, which is a key indicator for establishing loan size and interest rates.

Despite this task has often been evaluated through standard measurement tools (i.e., FICO[3] scores or credit and ratings for consumer and corporate, respectively), data-driven methodologies have been defined to deal with the credit scoring task (see Definition 41) harvesting data related to economic conditions and social-demographic information of the possible borrower.

Definition 41 Credit scoring Let $u \in U$ be a set of users, credit scoring is defined as the process to assign a dichotomous value (0/1) to a loan applicant u according to its risk in repaying the loan.

Nevertheless, one of the main issues in credit scoring concerns data unbalancing due to the lower number of defaulters users w.r.t. the normal ones, which implies a majority class bias.

4.2 Data preprocessing

In this section, we investigate how input data has been manipulated for improving the discriminatory power through different pre-processing strategies along three

[2] https://www.lendingclub.com

[3] https://www.fico.com/en/products/fico-score

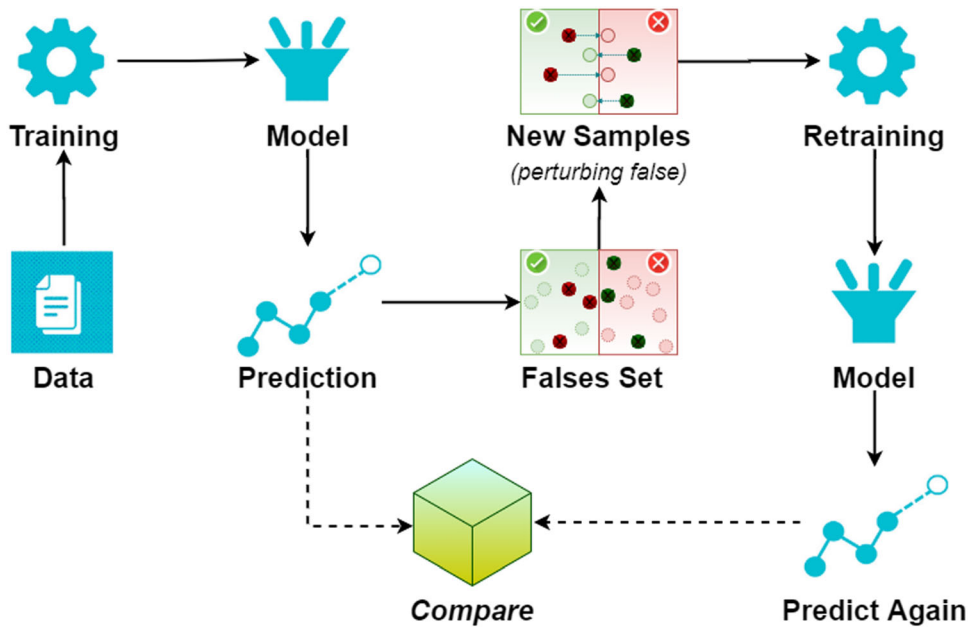

Fig. 1 Overview of the proposed methodology for credit risk scoring model. Data is provided as input to the model and predictions are compared to the ground truth to identify misclassified samples. These are utilized by CASTLE to generate a set of synthetic samples, which are added to the original dataset only if the prediction provided by the model is different from the initial prediction

main steps: *Cleaning*, *Transformation* and *Correlation analysis*. Herein, these operations become even more difficult because credit scoring relies on extremely unbalanced data [41]. We crawl data (about 2, 260, 668 samples with 145 features) from the *Lendingclub*[2] platform starting from 2007 until 2015, also considering current loans which are not expired. Before manipulating the dataset, we remove some features (i.e., *"post charge off gross recovery"* or *recovery*) that can be known only when the loan is finished [40]. Successively, we make pre-processing operations by filtering out features, whose null elements represent more than 80% of possible values together with all the rows containing at least one missing element. Furthermore, we only consider samples having *Fully paid*, *Default* and *Charged off* as loan status attribute. In particular, we deal with a binary classification problem by merging the last two values in the bad label. Data are, also, normalized by making a scaler operation defined in equation 1.

$$\text{scaled} = \frac{(\text{target - mean})}{stdDev} \tag{1}$$

We also compute the *Pearson correlation coefficient* for evaluating the relationship among the features, discarding the ones which have a coefficient value above 95%. The categorical features have been encoded through the *Binary encoder* for limiting the dimensionality explosion of the *One-hot encoding*, having some features with several possible values.

In conclusion, the filtered dataset is composed of 1, 136, 923 instances with 80 features, showing the ratio between good and bad loans around 4:1 according to the information represented in Table 1. Therefore, it is evident that the implementation of appropriate sampling techniques is crucial for the accurate analysis of these data.

Table 2 provides a summary of data processing actions with the related number of samples and features obtained at the end of each step.

Finally, the number of features has been diminished from 80 to 10, listed in Table 3, which has been selected on the basis of CASTLE's explanation. In particular, models trained on these 10 features have been compared w.r.t. the ones using the whole set of attributes obtaining small differences in terms of effectiveness performances. This outcome indicates that the top-ten attributes almost completely influence and explain predictions.

4.3 Classification

Credit scoring is an unbalanced and high-dimensional task relying on a large amount of borrower's historical information which contributes to compute the interest rate to be applied to the loan.

In this section, we discuss the classification module, which has been trained by splitting the cleaned data set into *training* and *test* set with a ratio 70:30 stratified on *loan status* attribute to avoid an unbiased evaluation. Thirty percent of the training dataset was retained as a validation set to select the best hyperparameters for the tested models.

Table 1 Dataset characterization

Class	Size
1 - Fully paid loans	638,365
0 - Defaulted loans	157,481

Table 2 Summary of data preprocessing actions

Rows	Cols	Action
2,260,668	145	Original dataset
2,260,668	49	Manually feature selection
2,260,668	40	Columns with too null values removal
2,002,890	40	Rows with missing values removal
1,136,923	40	Current loans removal
1,136,923	40	Low-variance features removal
1,136,923	38	Correlated features removal
1,136,923	80	Categorical features encoding

Table 3 Ten features identified by CASTLE framework

Attribute	Meaning
int_rate	Interest Rate on the loan
avg_cur_bal	Average current balance of all accounts
bc_open_to_buy	Total open to buy on revolving bankcards
revol_util	Revolving line utilization rate
revol_bal	Total credit revolving balance
total_bal_ex_mort	Total credit balance excluding mortgage
annual_inc	Borrower annual income self-reported
loan_amnt	The listed amount of the borrower's loan
total_acc	The total number of current credit lines
earliest_cr_line	Borrower's earliest reported credit line date

Once sampled the training set, the classifier's performances are assessed by varying the sampling techniques.

Different sampling techniques (random under sampling (RUS), random over sampling (ROS) and synthetic minority oversampling technique (SMOTE)) have been used for properly training each model by balancing samples into two classes (see Table 4). More in detail, the ROS creates novel samples by randomly duplicating the existing ones in the minority class while the RUS randomly filters samples from the majority class. In turn, SMOTE generates new samples in the minority class by computing the distance between the minority data and some selected samples through k-nearest neighbors.

Table 4 Dataset size according to different sampling strategies

Technique	Dataset size
RUS *(undersampling)*	314, 962
ROS *(oversampling)*	1, 276, 730
SMOTE *(oversampling)*	1, 276, 730

For improving the effectiveness performance of the classification model, our aim is to integrate CASTLE in the prediction process for supporting data augmentation. In particular, predictions are compared to the ground truth to identify misclassified samples, they are used by CASTLE to generate a set of synthetic samples, which will be added to the original dataset only if the prediction provided by the model is different from the initial prediction. Figure 2 summarizes the instances generation process in order to generate new instances for data augmentation. In particular, we perturb the misclassified predictions, whose uncertain probabilities are in the range of [0.4,0.6], by using the explanation vector, which represents the direction in which a sample needs to move for changing its prediction. Successively, we re-predict the instances and we select ones that become true that will be used for increasing the dataset size.

5 Evaluation

In the current section, we present an analysis of the proposed approach's effectiveness, as detailed in Sect. 5.1, by evaluating various classifiers commonly utilized in credit risk scoring, as outlined in previous studies [40, 41]. These classifiers are evaluated using various quantitative measures, as outlined in Sect. 5.2.

5.1 Experimental protocol

The objective of our analysis is to evaluate the effectiveness of the proposed methodology, outlined in Sect. 4, in comparison to various baselines trained with different sampling strategies, as well as state-of-the-art approaches and benchmarks. Specifically, we conduct three evaluations to assess the unique contributions of the proposed methodology:

- We evaluate the performance of all classifiers using three different sampling strategies (SMOTE, RUS, and ROS)
- We compare the results obtained by training all classifiers using our methodology
- We compare the results obtained to those of several state-of-the-art approaches ([54]) and benchmarks [40, 41].

Several classifiers have been considered according to the state-of-the-art for the credit risk [40, 41]: i) Decision tree, ii) Linear discriminant analysis (DT), iii) Random forest (RF) with 100 estimators, iv) Logistic regression (LR) maximum 200 interactions, v) AdaBoost (AB) with 100 estimators, vi) Stacking (ST) using as base-estimators *LR*,*LDA* and *RF* and an LR as a final estimator and the

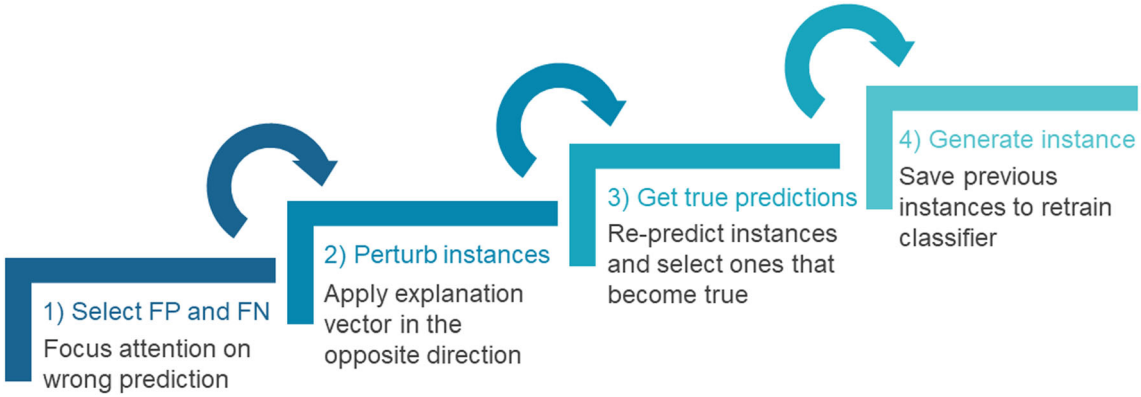

Fig. 2 Instances generation process

3-fold cross-validation, vii) Gradient boosting (GB) classifier, XGBoost (XGB).

The analysis has been performed on Google Colab,[4] a Platform-as-a-Service (PaaS) deployed on a virtual machine (VM) equipped with one single core XEON Processor @2.3 Ghz, 12 GB RAM and a Tesla K80 graphic processing unit (GPU). The software stack has been implemented in Python 3.6 and the baselines have been developed by scikit-learn,[5] also using *Imblearn*[6] for the sampling strategies.

5.2 Measure

Given that the dataset is highly imbalanced, relying solely on accuracy as a measure of performance may lead to misleading results due to the high number of positive predictions in the majority class. To mitigate this bias in the evaluation process, multiple performance measures are employed to assess the effectiveness of the proposed approach: Accuracy, FP-rate, Precision, Recall (a.k.a. Sensitivity), Specificity, G-Mean, and Area under the ROC Curve (AUC).

While the accuracy represents the number of correctly classified samples over the entire set of samples (see equation 2), the FP-rate is the ratio between the number of negative instances wrongly classified as positive and the total number of actual negative events, as shown in equation 3.

$$\text{Accuracy} = \frac{\text{TruePositive} + \text{FalsePositive}}{\text{TruePositive} + \text{FalsePositive} + \text{TrueNegative} + \text{FalseNegative}} \tag{2}$$

$$FP - \text{rate} = \frac{\text{FalsePositive}}{\text{FalsePositive} + \text{TrueNegative}} \tag{3}$$

Precision (Recall) is the fraction of retrieved relevant instances among the retrieved (relevant) instances as shown in equation 4 (5).

$$\text{Precision} = \frac{\text{TruePositive}}{\text{TruePositive} + \text{FalsePositive}} \tag{4}$$

$$\text{Recall(or Sensitivity)} = \frac{\text{TruePositive}}{\text{TruePositive} + \text{FalseNegative}} \tag{5}$$

Another metric is the Specificity (see equation 6) representing true negative proportion correctly identified by the model.

$$\text{Specificity} = \frac{\text{TrueNegative}}{\text{TrueNegative} + \text{FalsePositive}} \tag{6}$$

In turn, the geometric mean (7) measures the balance between classification performances in both the classes. A low G-Mean value represents a poor performance in positive prediction even if the negative cases are correctly classified.

$$G\text{ - Mean} = \sqrt{\text{Specificity*Sensitivity}} \tag{7}$$

Finally, the AUC measures the performance of a model by calculating the area under the ROC curve. The closer the area is to 1, the better the model is at distinguishing between the positive and negative classes.

6 Results

In this section, we discuss the obtained results according to the experimental protocol described in Sect. 5.1. In particular, we evaluate the effectiveness of our methodology in Sect. 6.1, also comparing it with respect to the best results obtained by applying different sampling strategies

[4] https://colab.research.google.com/
[5] https://scikit-learn.org/
[6] https://pypi.org/project/imblearn/

while Sect. 6.2 reports the comparison of our methodologies w.r.t. different baselines [40, 41] and [54].

6.1 Effectiveness of our methodology

Figure 3 presents the effectiveness of all classification models when utilizing RUS, ROS, and SMOTE sampling strategies, respectively. Within the domain of finance, the most significant metrics are the false positive rate (FP-rate) and the G-mean, as misclassification of default loans can result in higher costs compared to good loans, as highlighted in [20]. Therefore, our objective is to minimize the number of false positives, in order to reduce financial losses for the lenders.

As observed in Fig. 3, the SMOTE strategy is not an appropriate method for this specific case of imbalanced data. On the other hand, random undersampling demonstrates higher performance due to the ample amount of available data (exceeding 1.2 million samples). Furthermore, the SMOTE strategy results in a significant degradation of performance for all classifiers, with the exception of LDA, which remains unchanged in comparison to its undersampled version.

Although four classifiers (LDA, LR, AB, and GB) that are based on the ROS sampling strategy demonstrate superior performances when compared to their counterparts trained by RUS, in terms of G-Mean, the difference is relatively small, and the models can be considered approximately equal in terms of effectiveness. The other cases reveal the advantages of using the undersampling strategy. As a result, the classifier mostly predicts all instances as "good loan" resulting in high accuracy, due to the majority class size having a large number of samples as compared to the minority class, which affects its capability in predicting loan repayment.

Additionally, we evaluate the same classifiers on the entire dataset, which includes the 80 features filtered during data preprocessing, to compare the results obtained with the ones achieved by using the 10 features selected by LIME. Table 5 presents the results achieved through the undersampling strategy. The results of other sampling strategies are omitted as they did not show any improvement when compared to the counterparts trained on ten features.

In particular, the LDA, RF, LR, and ST models show an improvement of 2–5% in terms of FP-rate and G-mean measures, as they benefit from the increased amount of data. On the other hand, the other classifiers do not show significant improvement, with AB slightly deteriorating its performance. Since we observed an increase in effectiveness performance for a subset of classifiers with increasing complexity in terms of training time and inherent behavior, we consider the analysis performed on the dataset with only ten features, as the best outcomes are shown in Table 6.

Fig. 3 Classification performance of several classification models using random undersampling, random oversampling and SMOTE data augmentation strategies

Table 5 Classification performance of several classification models on the basis of all features using undersampling strategy

	Classifier	Accuracy	Recall	Precision	Specificity	FP-rate	G-mean	F1-score	AUC
Undersampling	DT	0.5738	0.8451	0.5738	0.5737	0.4263	0.6963	0.6835	0.5738
	LDA	0.6525	0.8877	0.6488	0.6673	0.3327	0.7697	0.7497	0.6580
	RF	0.6426	0.8910	0.6316	0.6870	0.3130	0.7824	0.7392	0.6593
	LR	0.6529	0.8882	0.6489	0.6688	0.3312	0.7707	0.7499	0.6589
	AB	0.6524	0.8915	0.6453	0.6815	0.3185	0.7795	0.7486	0.6634
	ST	0.6528	0.8889	0.6481	0.6717	0.3283	0.7727	0.7496	0.6599
	GB	0.6543	0.8925	0.6469	0.6841	0.3159	0.7814	0.7501	0.6655
	XGB	0.6563	0.8932	0.6491	0.6853	0.3147	0.7823	0.7519	0.6672

Table 6 Classifiers best performance with the related sampling strategy on the basis of ten features shown in Table 3

Sampling	Clf	Accuracy	Recall	Precision	Specificity	FP-rate	G-Mean	F1-score	AUC
Oversampling	AB	0.6235	0.8868	0.6083	0.6852	0.3148	0.7795	0.7216	0.6467
Oversampling	GB	0.6318	0.8856	0.6212	0.6748	0.3252	0.7731	0.7302	0.6480
Undersampling	XGB	0.6367	0.8866	0.6272	0.6748	0.3252	0.7735	0.7216	0.640
Undersampling	RF	0.6222	0.8823	0.6105	0.6697	0.3303	0.7687	0.7216	0.6401
Undersampling	ST	0.6457	0.8795	0.6470	0.6406	0.3595	0.7506	0.7455	0.6438
Oversampling	LR	0.6527	0.8764	0.6601	0.6227	0.3773	0.7389	0.7531	0.6414
Undersampling	DT	0.5624	0.8388	0.5626	0.5617	0.4383	0.6864	0.6735	0.5621
Oversampling	LDA	0.6562	0.8753	0.6663	0.6153	0.3847	0.7339	0.7566	0.6408

We applied our methodology to this dataset to compare the obtained results and to highlight the key differences with traditional approaches. First, we perturb misclassified instances to alter the decision region by following the opposite prediction direction provided by *CASTLE*, and we, subsequently, iterate the process until the required number of synthetic samples is reached. In particular, we explain 2000 target instances for each classifier, which were previously misjudged by the chosen black-box model. Then, for each individual instance, a fixed number of perturbations is applied to generate about 150,000 synthetic samples. By adjusting the parameters, this number fluctuates within a range of 140 and 160 thousand new samples.

The set of synthetic instances is fed back to each classifier multiple times, varying the number of retraining samples each time. Their size is progressively changed, starting from ten thousand samples, adding another 10 thousand samples each time up to the maximum number of instances generated for the specific classifier. We compared the effectiveness of the retrained classifiers with the best results obtained previously, using the sampling technique that enabled maximum performance. The best classifiers among those analyzed are ST and AB, specifically for recognizing defaulted loans, or GB and XGB if a balance between accuracy and false recognition is sought. The performance of RF remains almost unchanged, while a significant improvement has been observed in LR, whose performance has increased by 12% in the FP-rate. However, LDA is the only model that shows a deterioration in its performance.

6.2 Comparison w.r.t. baselines

In this section, we compare the obtained results with different state-of-the-art benchmarks ([40, 41]) and approaches ([54]), see Table 7.

Although our best results achieve lower accuracy scores than those reported in [41] and [40], our approach achieves lower FP-Rate (0.230) and higher G-Mean (0.841) compared to the best results in both benchmarks. Our classifiers generally show improved performance in G-Mean and FP-Rate compared to most baseline methods, indicating better handling of class imbalance and fewer false positives. The obtained outcome is particularly relevant for credit scoring tasks, as misclassification of default loss incurs higher costs than those related to good loans, as detailed in [20]. On the other hand, the best results in [54] show higher Specificity values compared to our results and those of the

Table 7 Comparison with baselines. OS and US state for oversampling and undersampling, respectively

Baselines	Classifier	Accuracy	FP-Rate	G-Mean	AUC
[40]	RF-RUS	0.717	0.320	0.6560	0.640
	LR-ROS	0.710	0.360	0.6503	0.656
	LR-smootetoken	0.710	0.360	0.6500	0.656
	Logistic regression	0.685	0.960	0.2600	0.770
	Random forest	0.720	0.920	0.2870	0.773
	MLP	0.704	0.945	0.2060	0.771
[41]	RF-RUS	0.696	0.420	0.650	0.6920
	LDA-SMOTE	0.700	0.350	0.643	0.6400
	LR-smootetoken	0.700	0.352	0.643	0.6400
	Logistic regression	0.703	0.950	0.218	0.8173
	Random forest	0.696	0.980	0.12	0.8176
[54]	[54]	0.7231		0.6009	0.6697
	GDBT (OS)	0.6235	-	0.6207	0.6207
	Random forest (OS)	0.7701	-	0.5134	0.5795
	AdaBoost (OS)	0.7562	-	0.4050	0.5224
	Decision tree (OS)	0.7568	-	0.4060	0.5231
	Logistic regression (OS)	0.5630	-	0.5597	0.5600
	Multilayer perceptron (OS)	0.7245	-	0.3593	0.4892
	GDBT (US)	0.6033	-	0.6138	0.6140
	Random forest (US)	0.5912	-	0.6193	0.6207
	AdaBoost (US)	0.5288	-	0.5404	0.5408
	Decision tree (US)	0.5323	-	0.5418	0.5421
	Logistic regression (US)	0.5742	-	0.5609	0.5615
	Multilayer perceptron (US)	0.7245	-	0.3593	0.4892
Ours	Decision tree	0.563	0.432	0.691	0.565
	LDA	0.669	0.469	0.680	0.619
	Random forest	0.621	0.327	0.771	0.641
	Logistic regression	0.578	0.289	0.798	0.629
	AdaBoost	0.573	0.276	0.803	0.630
	Stochastic trees	0.533	0.230	0.841	0.624
	Gradient boost	0.612	0.303	0.787	0.644
	XGBoost	0.631	0.312	0.782	0.653

benchmarks, however, both Sensitivity and G-Mean scores are much lower than ours.

7 Conclusion

In this paper, we integrate the *CASTLE* XAI framework, a method to explain black-box classification models, in the credit scoring task with the aim to support the data augmentation process. Dealing with an imbalance problem, *CASTLE* has been proposed as a new approach to generate synthetic samples in order to retrain models through explanations, which has powered better performance than traditional models in the literature. The analysis has been made on the largest social lending platform (*Lending club*), which has been the first peer-to-peer lender to register its offerings as securities and to offer loan trading on a secondary market. Overall, we discovered that generating new instances along the decision boundary of the classification model always benefits the performance of the credit score model.

We have planned as future works to apply this novel methodology to other real Social Lending platforms as well as other application domains, also investigating their feasibility to more recent approaches (i.e., deep learning or ensemble strategies).

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Declarations

Conflict of interest The authors declare that they have no conflict of interest.

Ethics approval This article does not contain any studies with human participants performed by any of the authors.

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References

1. Basha AS, Elgammal MM, Abuzayed BM (2021) Online peer-to-peer lending: a review of the literature. Electron Commer Res Appl 48:101069. https://doi.org/10.1016/j.elerap.2021.101069
2. Abdoli M, Akbari M, Shahrabi J (2023) Bagging supervised autoencoder classifier for credit scoring. Expert Syst Appl 213:118991. https://doi.org/10.1016/j.eswa.2022.118991
3. Ashofteh A, Bravo JM (2021) A conservative approach for online credit scoring. Expert Syst Appl 176:114835. https://doi.org/10.1016/j.eswa.2021.114835
4. Baltes S, Ralph P (2022) Sampling in software engineering research: a critical review and guidelines. Empir Softw Eng 27(4):94. https://doi.org/10.1007/s10664-021-10072-8
5. Balyuk T (2016) Financial innovation and borrowers: Evidence from peer-to-peer lending. University of Toronto-Rotman School of Management
6. Barocas S, Hardt M, Narayanan A (2019) Fairness in machine learning. fairmlbook.org
7. Bastani K, Asgari E, Namavari H (2019) Wide and deep learning for peer-to-peer lending. Expert Syst Appl 134:209–224. https://doi.org/10.1016/j.eswa.2019.05.042
8. Bussmann N, Giudici P, Marinelli D, Papenbrock J (2021) Explainable machine learning in credit risk management. Comput Econ 57(1):203–216
9. Cao L (2022) AI in finance: challenges, techniques, and opportunities. ACM Comput Surv 55(3):1–38. https://doi.org/10.1145/3502289
10. Carcillo F, Le Borgne YA, Caelen O, Kessaci Y, Oblé F, Bontempi G (2021) Combining unsupervised and supervised learning in credit card fraud detection. Inf Sci 557:317–331. https://doi.org/10.1016/j.ins.2019.05.042
11. Carta S, Ferreira A, Reforgiato Recupero D, Saia R (2021) Credit scoring by leveraging an ensemble stochastic criterion in a transformed feature space. Prog Artif Intell 10(4):417–432. https://doi.org/10.1007/s13748-021-00246-2
12. Chen C, Lin K, Rudin C, Shaposhnik Y, Wang S, Wang T (2022) A holistic approach to interpretability in financial lending: models, visualizations, and summary-explanations. Decis Support Syst 152:113647. https://doi.org/10.1016/j.dss.2021.113647
13. Dastile X, Celik T, Potsane M (2020) Statistical and machine learning models in credit scoring: a systematic literature survey. Appl Soft Comput 91:106263. https://doi.org/10.1016/j.asoc.2020.106263
14. Diamond DW, Dybvig PH (1983) Bank runs, deposit insurance, and liquidity. J Polit Econ 91(3):401–419. https://doi.org/10.1086/261155
15. Djeundje VB, Crook J, Calabrese R, Hamid M (2021) Enhancing credit scoring with alternative data. Expert Syst Appl 163:113766. https://doi.org/10.1016/j.eswa.2020.113766
16. Emekter R, Tu Y, Jirasakuldech B, Lu M (2015) Evaluating credit risk and loan performance in online Peer-to-Peer (P2P) lending. Appl Econ 47(1):54–70. https://doi.org/10.1080/00036846.2014.962222
17. Fiore U, De Santis A, Perla F, Zanetti P, Palmieri F (2019) Using generative adversarial networks for improving classification effectiveness in credit card fraud detection. Inf Sci 479:448–455. https://doi.org/10.1016/j.ins.2017.12.030
18. Foo J, Lim L H, Wong K SW (2017) Macroeconomics and FinTech: Uncovering latent macroeconomic effects on peer-to-peer lending. arXiv preprint arXiv:1710.11283
19. Gao GX, Fan ZP, Fang X, Lim YF (2018) Optimal Stackelberg strategies for financing a supply chain through online peer-to-peer lending. Eur J Oper Res 267(2):585–597. https://doi.org/10.1016/j.ejor.2017.12.006
20. García V, Marqués AI, Sánchez JS (2019) Exploring the synergetic effects of sample types on the performance of ensembles for credit risk and corporate bankruptcy prediction. Inform Fusion 47:88–101. https://doi.org/10.1016/j.inffus.2018.07.004
21. Han L, Chen T, Demartini G, Indulska M, Sadiq S (2023) A data-driven analysis of behaviors in data curation processes. ACM Trans Inform Syst 41(3):1–35. https://doi.org/10.1145/3567419
22. Han L, Rajasekar A, Li S (2022) An evidence-based credit evaluation ensemble framework for online retail smes. Knowl Inf Syst 64(6):1603–1623. https://doi.org/10.1007/s10115-022-01682-9
23. Hayashi Y (2016) Application of a rule extraction algorithm family based on the Re-RX algorithm to financial credit risk assessment from a Pareto optimal perspective. Oper Res Persp 3:32–42. https://doi.org/10.1016/j.orp.2016.08.001
24. Hui Hou W, Kang Wang X, Yu Zhang H, Giang Wang J, Li L (2020) A novel dynamic ensemble selection classifier for an imbalanced data set: an application for credit risk assessment. Knowl-Based Syst 208:106462
25. Kaur H, Pannu HS, Malhi AK (2019) A systematic review on imbalanced data challenges in machine learning: applications and solutions. ACM Comput. Surv. 52(4):1–36. https://doi.org/10.1145/3343440
26. Kim A, Cho SB (2019) An ensemble semi-supervised learning method for predicting defaults in social lending. Eng Appl Artif Intell 81:193–199. https://doi.org/10.1016/j.engappai.2019.02.014
27. Kozodoi N, Jacob J, Lessmann S (2022) Fairness in credit scoring: assessment, implementation and profit implications. Eur J

Oper Res 297(3):1083–1094. https://doi.org/10.1016/j.ejor.2021.06.023
28. Kozodoi N, Lessmann S, Baesens B, Papakonstantinou K (2019) Profit-Oriented Feature Selection in Credit Scoring Applications. In: Operations Research Proceedings 2018: Selected Papers of the Annual International Conference of the German Operations Research Society (GOR), Brussels, Belgium, September 12-14, 2018, pp. 59–65. Springer https://doi.org/10.1007/978-3-030-18500-8_9
29. La Gatta V, Moscato V, Postiglione M, Sperlì G (2021) CASTLE: cluster-aided space transformation for local explanations. Expert Syst Appl 179:115045. https://doi.org/10.1016/j.eswa.2021.115045
30. Lan Q, Jiang S (2021) A method of credit evaluation modeling based on block-wise missing data. Appl Intell 51(10):6859–6880. https://doi.org/10.1007/s10489-021-02225-5
31. Lewy D, Mańdziuk J (2023) An overview of mixing augmentation methods and augmentation strategies. Artif Intell Rev 56(3):2111–2169. https://doi.org/10.1007/s10462-022-10227-z
32. Li H, Cao Y, Li S, Zhao J, Sun Y (2020) XGBoost model and its application to personal credit evaluation. IEEE Intell Syst 35(3):52–61. https://doi.org/10.1109/MIS.2020.2972533
33. Li Z, Zhang J, Yao X, Kou G (2021) How to identify early defaults in online lending: a cost-sensitive multi-layer learning framework. Knowl-Based Syst 221:106963. https://doi.org/10.1016/j.knosys.2021.106963
34. Liu J, Zhang S, Fan H (2022) A two-stage hybrid credit default evaluation model using XGBoost and graph-based deep neural network. Exp Syst Appl. https://doi.org/10.1016/j.eswa.2022.116624
35. Liu Z, Shang J, Wu Sy, Chen Py (2020) Social collateral, soft information and online peer-to-peer lending: a theoretical model. Eur J Oper Res 281(2):428–438. https://doi.org/10.1016/j.ejor.2019.08.038
36. Louzada F, Ara A, Fernandes GB (2016) Classification methods applied to credit scoring: systematic review and overall comparison. Surv Oper Res Manag Sci 21(2):117–134. https://doi.org/10.1016/j.sorms.2016.10.001
37. Pricing mechanisms in the online Peer-to-Peer lending market (2017) jiang Ma, B., long Zhou, Z., ying Hu, F. Electron Commer Res Appl 26:119–130. https://doi.org/10.1016/j.elerap.2017.10.006
38. Ma X, Sha J, Wang D, Yu Y, Yang Q, Niu X (2018) Study on a prediction of P2P network loan default based on the machine learning LightGBM and XGboost algorithms according to different high dimensional data cleaning. Electron Commer Res Appl 31:24–39. https://doi.org/10.1016/j.elerap.2018.08.002
39. Malekipirbazari M, Aksakalli V (2015) Risk assessment in social lending via random forests. Expert Syst Appl 42(10):4621–4631. https://doi.org/10.1016/j.eswa.2015.02.001
40. Moscato V, Picariello A, Sperlí G (2021) A benchmark of machine learning approaches for credit score prediction. Expert Syst Appl 165:113986. https://doi.org/10.1016/j.eswa.2020.113986
41. Namvar A, Siami M, Rabhi F, Naderpour M (2018) Credit risk prediction in an imbalanced social lending environment. Int J Comput Intell Syst 11(1):925–935. https://doi.org/10.2991/ijcis.11.1.70
42. Philippon T (2016) The fintech opportunity. Tech. rep, National Bureau of Economic Research
43. Qi Y, Xiao J (2018) Fintech: AI powers financial services to improve people's lives. Commun ACM 61(11):65–69. https://doi.org/10.1145/3239550
44. Qian X, Cai HH, Innab N, Wang D, Ciano T, Ahmadian A (2024) A novel deep learning approach to enhance creditworthiness evaluation and ethical lending practices in the economy. Annal Oper Res. https://doi.org/10.1007/s10479-024-05849-1
45. Ribeiro M T, Singh S, Guestrin C (2016) "Why Should I Trust You?": Explaining the Predictions of Any Classifier. In: Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, KDD '16, p. 1135–1144. Association for Computing Machinery, New York, NY, USA https://doi.org/10.1145/2939672.2939778
46. Roeder J, Palmer M, Muntermann J (2022) Data-driven decision-making in credit risk management: the information value of analyst reports. Decis Support Syst 158:113770. https://doi.org/10.1016/j.dss.2022.113770
47. Rozo BJG, Crook J, Andreeva G (2023) The role of web browsing in credit risk prediction. Decis Support Syst 164:113879. https://doi.org/10.1016/j.dss.2022.113879
48. Rudin C (2019) Stop explaining black box machine learning models for high stakes decisions and use interpretable models instead. Nat Mach Intell 1(5):206–215. https://doi.org/10.1038/s42256-019-0048-x
49. Runchi Z, Liguo X, Qin W (2023) An ensemble credit scoring model based on logistic regression with heterogeneous balancing and weighting effects. Expert Syst Appl 212:118732. https://doi.org/10.1016/j.eswa.2022.118732
50. Saia R, Carta S, Recupero D R, Fenu G, Saia M (2019) A Discretized Enriched Technique to Enhance Machine Learning Performance in Credit Scoring. In: KDIR, pp. 202–213 https://doi.org/10.5220/0008377702020213
51. Saini M, Susan S (2023) Tackling class imbalance in computer vision: a contemporary review. Artif Intell Rev 56(Suppl 1):1279–1335. https://doi.org/10.1007/s10462-023-10557-6
52. Santoso W, Trinugroho I, Risfandy T (2020) What determine loan rate and default status in financial technology online direct lending? Evidence from Indonesia. Emerg Mark Financ Trade 56(2):351–369. https://doi.org/10.1080/1540496X.2019.1605595
53. Simumba N, Okami S, Kodaka A, Kohtake N (2022) Multiple objective metaheuristics for feature selection based on stakeholder requirements in credit scoring. Decis Support Syst 155:113714. https://doi.org/10.1016/j.dss.2021.113714
54. Song Y, Wang Y, Ye X, Wang D, Yin Y, Wang Y (2020) Multi-view ensemble learning based on distance-to-model and adaptive clustering for imbalanced credit risk assessment in P2P lending. Inf Sci 525:182–204. https://doi.org/10.1016/j.ins.2020.03.027
55. Soui M, Gasmi I, Smiti S, Ghédira K (2019) Rule-based credit risk assessment model using multi-objective evolutionary algorithms. Expert Syst Appl 126:144–157. https://doi.org/10.1016/j.eswa.2019.01.078
56. Talaat FM, Aljadani A, Badawy M, Elhosseini M (2023) Toward interpretable credit scoring: integrating explainable artificial intelligence with deep learning for credit card default prediction. Neural Comput Appl. https://doi.org/10.1007/s00521-023-09232-2
57. Teles G, Rodrigues JJ, Saleem K, Kozlov S, Rabêlo RA (2020) Machine learning and decision support system on credit scoring. Neural Comput Appl 32:9809–9826. https://doi.org/10.1007/s00521-019-04537-7

58. Teply P, Polena M (2020) Best classification algorithms in peer-to-peer lending. Nor Am J Econ Financ 51:100904. https://doi.org/10.1016/j.najef.2019.01.001
59. Xiao J, Zhou X, Zhong Y, Xie L, Gu X, Liu D (2020) Cost-sensitive semi-supervised selective ensemble model for customer credit scoring. Knowl-Based Syst 189:105118. https://doi.org/10.1016/j.knosys.2019.105118
60. Ye X, an Dong L, Ma D (2018) Loan evaluation in P2P lending based on random forest optimized by genetic algorithm with profit score. Electron Commer Res Appl 32:23–36. https://doi.org/10.1016/j.elerap.2018.10.004
61. Zhang W, Yang D, Zhang S (2021) A new hybrid ensemble model with voting-based outlier detection and balanced sampling for credit scoring. Expert Syst Appl 174:114744. https://doi.org/10.1016/j.eswa.2021.114744
62. Zhen H, Xiong S, Gong W, Wang L (2023) Neighborhood evolutionary sampling with dynamic repulsion for expensive multimodal optimization. Inf Sci 630:82–97. https://doi.org/10.1016/j.ins.2023.02.049
63. Zhou L, Fujita H, Ding H, Ma R (2021) Credit risk modeling on data with two timestamps in peer-to-peer lending by gradient boosting. Appl Soft Comput 110:107672. https://doi.org/10.1016/j.asoc.2021.107672

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